DOMAIN 8: MULTI-AGENT AI TELEMETRY, REASONING TRACES & AUDITING

7 Core Tables • LangGraph 35-Field State Audit, LLM Token Telemetry, Knowledge Documents & Alerts

7 TABLES

agent_definitions — Specialist Agent Directory

PK	**agent_id**	UUID
	agent_code	VARCHAR
	agent_name	VARCHAR
	purpose	VARCHAR
	is_active	BOOLEAN
	created_at	TIMESTAMP

agent_versions — Prompt & Model Checkpoints

PK	**agent_version_id**	UUID
FK	**agent_id**	UUID
	version_label	VARCHAR
	llm_provider	VARCHAR
	model_name	VARCHAR
	prompt_version	VARCHAR
	prompt_hash	VARCHAR
	structured_output_schema_version	VARCHAR
	configuration	JSONB
	is_active	BOOLEAN
	created_at	TIMESTAMP

agent_execution_runs — State Graph Execution Traces

PK	**agent_execution_run_id**	UUID
FK	**generation_request_id**	UUID
FK	**portfolio_version_id**	UUID
FK	**agent_version_id**	UUID
	execution_status	VARCHAR
	input_fingerprint	VARCHAR
	output_summary	JSONB
	input_token_count	INTEGER
	output_token_count	INTEGER
	latency_ms	INTEGER
	error_code	VARCHAR
	error_summary	VARCHAR
	started_at	TIMESTAMP
	completed_at	TIMESTAMP

knowledge_documents — Indexed Regulatory Grounding

PK	**knowledge_document_id**	UUID
FK	**data_source_id**	UUID
	external_document_id	VARCHAR
	document_title	VARCHAR
	document_type	VARCHAR
	source_url	VARCHAR
	content_hash	VARCHAR
	object_storage_uri	VARCHAR
	embedding_model	VARCHAR
	vector_index_reference	VARCHAR
	index_status	VARCHAR
	published_at	TIMESTAMP
	indexed_at	TIMESTAMP
	metadata	JSONB
	created_at	TIMESTAMP

audit_events — Security Access Log

PK	**audit_event_id**	UUID
FK	**actor_user_id**	UUID
	action_code	VARCHAR
	resource_type	VARCHAR
	resource_id	UUID
	outcome	VARCHAR
	request_id	VARCHAR
	ip_address	VARCHAR
	user_agent	VARCHAR
	event_metadata	JSONB
	created_at	TIMESTAMP

notifications — User Rebalancing Alerts

PK	**notification_id**	UUID
FK	**user_id**	UUID
	notification_type	VARCHAR
	title	VARCHAR
	message	VARCHAR
	delivery_channel	VARCHAR
	delivery_status	VARCHAR
	created_at	TIMESTAMP
	sent_at	TIMESTAMP
	read_at	TIMESTAMP
	dismissed_at	TIMESTAMP
	metadata	JSONB

user_feedback — Recommendation Ratings

PK	**user_feedback_id**	UUID
FK	**user_id**	UUID
FK	**portfolio_version_id**	UUID
	feedback_type	VARCHAR
	rating	INTEGER
	feedback_text	VARCHAR
	created_at	TIMESTAMP